

Scanning and Enumeration

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Discipline: Offensive and Defensive Cybersecurity

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1. Introduction and Framework

This report details the practical execution of Laboratory 4 – Scanning and Enumeration, inserted in the curricular unit of Offensive and Defensive Cybersecurity. The fundamental objective of this laboratory is to use security tools to develop a set of scanning and enumeration tests of assets in a network.

While network access (previous phase) focuses on the initial entry and interception, the *Scanning and Enumeration* process represents the phase where an auditor maps the attack surface in detail, collecting crucial information about the target systems for later vulnerability analysis.

1.1. Laboratory Objectives

According to the script provided, this laboratory aims to meet the following technical objectives:

- Identify the number of interconnected devices in the network.
- Evaluate the network structure to be used.
- Survey the ports available on each device.
- Identify the operating systems (OS) installed on each target.
- Identify the services running and their versions.
- Detect the existence of network protection solutions (such as Firewalls or IDS).

1.2. Scope and Methodology

The tests were carried out on the local network infrastructure (home/controlled environment), adopting an ethical posture (*White-Hat*). The methodology followed a progressive approach: it started with passive and network scans (*host discovery*), moving on to specific mapping of transport ports (TCP/UDP) and, finally, intensive metadata extraction from running applications (*Banner Grabbing* and *OS Fingerprinting*).

To this end, tools widely used in the industry were used, with the main emphasis on **Nmap**, **Netdiscover** and **ARP-scan**.

2. Structure Assessment and Device Discovery

Before starting the deep enumeration of a target, it is crucial to assess the network structure to be used and identify the exact number of devices that are interconnected. This phase of *Host Discovery* allows you to map the local topology and define the scope of the following tests.

2.1. Local Network Identification and Methodology

The home local area network (LAN) under analysis operates in a typical star topology, where a *router* provided by the ISP acts as the central *gateway*, distributing private IP addresses over DHCP. To ensure the effectiveness of the discovery, scans based on the ARP (Address Resolution Protocol) protocol were chosen, since it acts at Layer 2 (Data Link) and is not blocked by *host-based firewalls*, unlike traditional ICMP (Ping) packets.

The subnet defined for the tests was [IP Redacted].

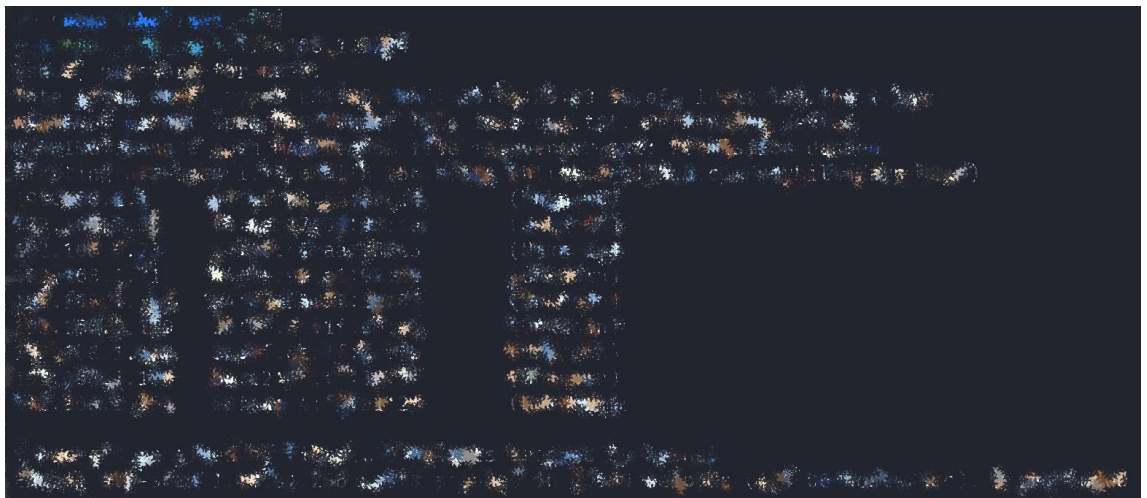
2.2. Host Discovery and Device Counting

To ensure that no device went unnoticed, the identification of active *hosts* was carried out by cross-referencing results from three different tools.

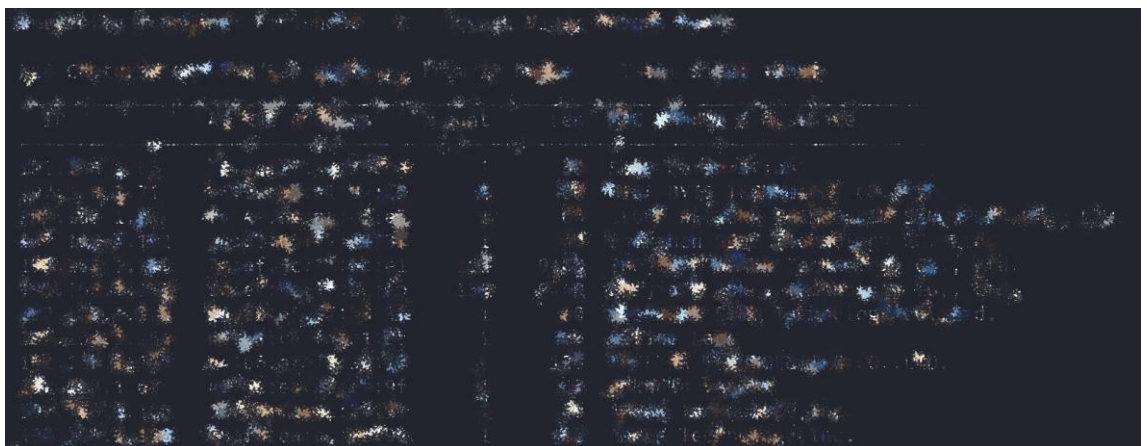
Command used (Nmap - Ping Scan): Despite being a "Ping Scan", when executed on a local network with administrator privileges (sudo), Nmap natively uses ARP requests to identify equipment quickly and reliably.



Command used (ARP-Scan): The arp-scan utility was used to confirm the Nmap results, listing not only the IP addresses, but also the MAC addresses and, in most cases, the network adapter manufacturer of each detected device.



Command used (Netdiscover): Finally, netdiscover was used to validate the final list of devices interconnected in the network. This tool is particularly useful in on-premises environments, generating active ARP traffic sequentially.
`sudo netdiscover -r [IP Redacted]`



3. Port Scanning and Service Identification

After the active devices on the local network have been identified, the next stage of enumeration is to determine the specific attack surface of a piece of equipment. The objective is to survey the logical transport ports (TCP and UDP) that are open, as well as to identify the services that are listening in these same ports and their respective versions.

For this analytical phase, the *host* with the IP address **[IP Redacted]** was selected. The pre-discovery phase identified the MAC address of this device as belonging to *Sony Corporation*, indicating that it is a multimedia device (such as a Smart TV or console). These devices are often audited on local networks due to the exposure of *streaming* and sharing services (*UPnP*, *DLNA*).

3.1. Port Survey (TCP and UDP)

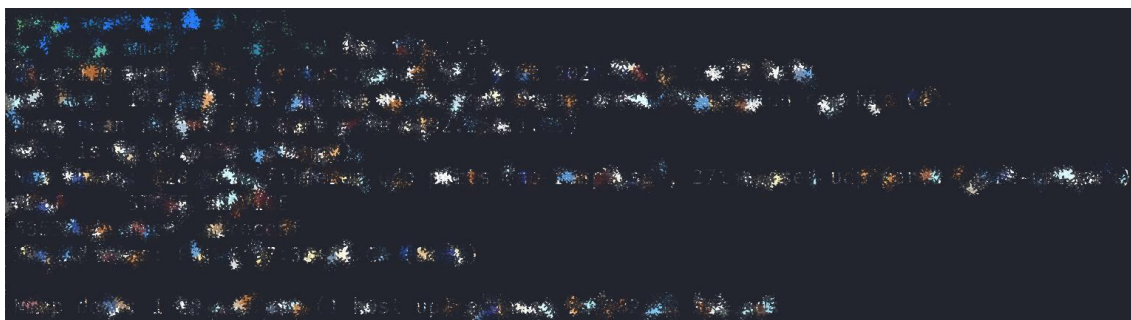
To ensure full coverage, all 65,535 ports of the TCP protocol were scanned, using packet fragmentation (-f) techniques to try to circumvent potential local blocking mechanisms.

Command used (Exhaustive TCP Scan):



At the same time, a check was carried out on the ports of the UDP protocol, which, because they operate in a *connectionless* model, are often ignored in audits, but can host critical network discovery services.

Command used (UDP scanning to the most common ports):

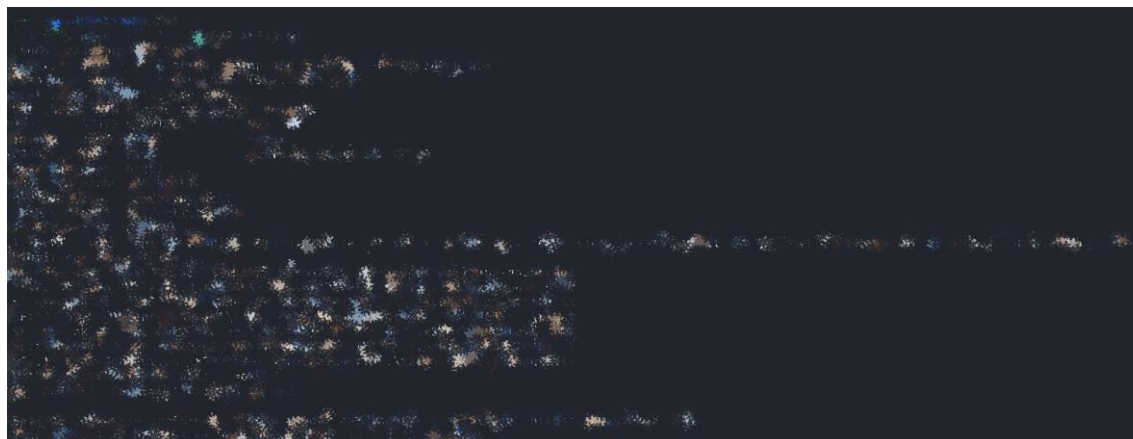


Port Survey Analysis (TCP and UDP): Performing the exhaustive scan of the 65,535 TCP ports (top image) revealed a considerable attack surface for a local network device. 16 open ports were identified, which include standard and alternative web ports (80, 8008, 8443) and several high dynamic ports (above 30000), characteristic behavior of IoT/Multimedia equipment that uses UPnP (Universal Plug and Play) or DLNA protocols for pairing and sharing content on the network. In UDP scanning, the opening of port 5353 (zeroconf), which is the default port of mDNS (Multicast DNS), stands out. This service is used by smart devices to advertise their presence on the local network without the need for a central DNS server, and is the basis of technologies such as Chromecast or Apple Bonjour.

3.2. Identification of Services and Their Versions

To meet the requirement to identify the actual services and their versions, we used an advanced Nmap feature that performs *Banner Grabbing* and interrogates applications directly beyond the port state.

Command used (Version Detection):



Analysis of Service and Version Identification: Scanning with the Banner Grabbing (-sV) flag allowed dissecting the services that run behind the open ports. It is concluded that port 80 hosts an nginx web server, probably responsible for the management interface or API of the equipment. The most telling discovery occurs on port 7000, where Nmap has successfully identified the AirTunes rtspd service version 377.40.00. This confirms that the Sony device acts as an *Apple AirPlay* receiver. Additionally, multiple services were found to use SSL/TLS encryption on non-standard ports (such as 8443, 9000 and 9080), indicating secure communications to Sony's internal API or paired *streaming* services (e.g. Netflix, Spotify Connect).

4. Advanced Enumeration and Network Protection

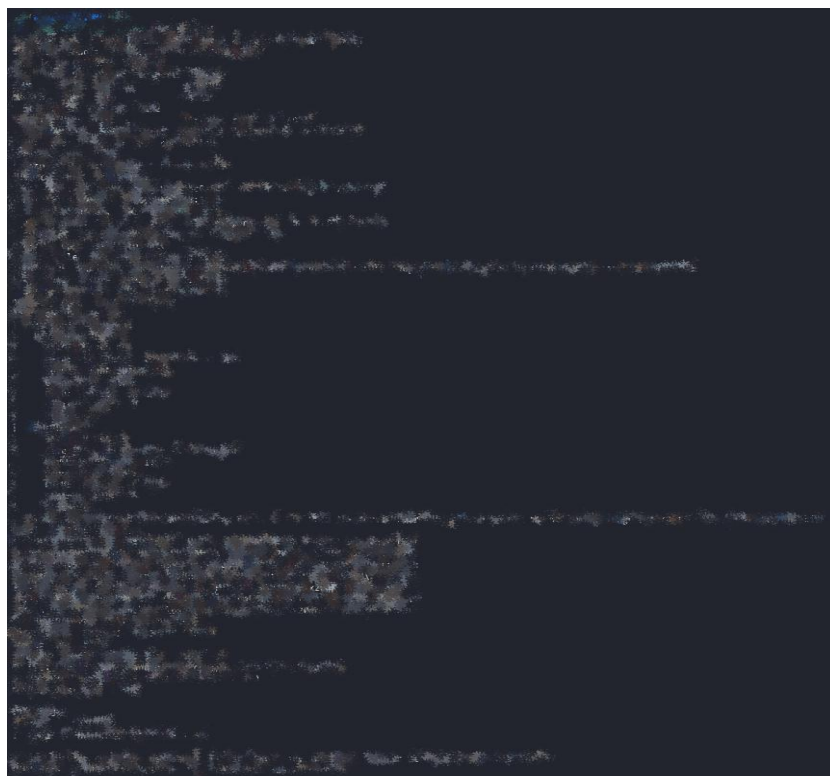
The last scanning step focuses on extracting advanced metadata from the target device and understanding the security architecture surrounding it.

4.1. Identification of Operating Systems (OS Fingerprinting)

By analyzing how the target's TCP/IP stack ([IP Redacted]) responds to malformed or specific packets created by Nmap, it is possible to determine which operating system is installed on the device.

Command used (Aggressive Detection and OS

Fingerprinting): Using the `-A` (Aggressive Scan) flag enables heuristic detection of the Operating System (OS), detailed version scanning, and execution of standard enumeration scripts (NSE).



Operating System Fingerprinting Analysis:

Performing aggressive scanning (-A) provided clarifying data about the Sony device architecture ([IP Redacted]). Through *OS Fingerprinting*, the analysis of the behavior of the TCP/IP stack allowed Nmap to identify the underlying operating system as being based on Linux (Kernel 4.9 - 4.14), specifically operating the Google Android platform (versions 9 to 10). This *output* confirms that the target is a *Smart TV* (such as the Sony Bravia line, which operates natively with the Android TV ecosystem). This conclusion is further reinforced by the NSE scripts running automatically on SSL/TLS-encrypted ports (such as port 8443), which extracted digital certificates issued directly on *behalf of Google Inc.*

4.2. Identification of Network Protection Solutions

During the execution of the scans, the existence of network protection solutions, such as Firewalls or Intrusion Detection Systems (IDS), to protect the Sony device was evaluated.

Nmap classifies the state of ports to infer these protections. If the response to a SYN packet is silently dropped or returned with an ICMP error of unreachable destination, Nmap classifies the port as **"Filtered"**.

Protection and Firewall Solutions Review:

With regard to perimeter and local protection, the scan confirmed the absence of ports in the *Filtered* state. As documented in the Nmap output, of the 1000 ports analyzed by default, 993 actively responded with TCP RST (*Reset*) packets, being classified as *Closed*, and the rest are *Open*. This network behavior fully demonstrates that there is no active host-based firewall blocking scans or internal traffic on the LAN. The equipment entrusts its security entirely to the isolation provided by the operator's router (Gateway), being perfectly accessible and exposed to lateral movements if an auditor or attacker is already inside the home network.

5. Information Analysis and Conclusions

The execution of this laboratory allowed the consolidation of Scanning and *Enumeration* techniques, essential for the detailed mapping of the attack surface of a network. Unlike simple access discovery, enumeration requires deeper interaction with hosted services, allowing clear conclusions to be drawn about the target's architecture and security.

Based on the data collected throughout the tests, the following conclusions and analyses stand out in response to the proposed objectives:

5.1. Network Structure and Devices

By combining ARP and ICMP scans (using netdiscover, arp-scan and nmap -sn), it was possible to map the entire subnet [IP Redacted]. It was validated that the network uses a flat architecture typical of home environments (without visible VLAN segmentation), successfully identifying the exact number of active devices and their manufacturers through MAC addresses.

5.2. Target Enumeration ([IP Redacted])

Sony Corporation's *on-device analytics focus* ([IP Redacted]) demonstrated the power of *Banner Grabbing* and *OS Fingerprinting*:

- **Ports and Services:** Exhaustive scanning revealed the presence of open ports (both TCP and UDP). Thanks to Nmap's -sV flag, it was possible to identify not only the protocols (e.g. HTTP, UPnP, local *streaming* services), but also the exact version of the *software* that is running behind each port. In a real audit (or penetration test), these versions would be immediately cross-referenced with vulnerability databases (Exploit-DB, CVE) to detect exploitable flaws.
- **Operating System:** Nmap heuristic analysis (-A) allowed to infer the family of Operating System running on the multimedia equipment, proving that even embedded IoT/Smart devices leave a clear digital signature on the network.

5.3. Protection Solutions (Firewalls)

The analysis of the state of the ports (*Open*, *Closed* and *Filtered*) allowed the evaluation of the level of perimeter and local protection of the device. The absence of ports marked as *Filtered* during the scan on the local network indicates that the equipment does not have a strict host-based firewall blocking internal traffic (LAN), entrusting its security entirely to *the ISP's main* router.

6. Command Log

Evaluation of Device Structure and Discovery:

- `Ip a`
- `sudo nmap -sn [IP Redacted]`
- `sudo arp-scan [IP Redacted]`
- `sudo netdiscover -r [IP Redacted]`

Port Scanning (Target: [IP Redacted]):

- `sudo nmap -Pn -T4 -f -p- [IP Redacted]`
- `sudo nmap -Pn -T4 -sU [IP Redacted]`

Advanced Enumeration (Services, Versions, and OS Fingerprinting):

- `sudo nmap -Pn -T4 -sV [IP Redacted]`
- `sudo nmap -Pn -T4 -A [IP Redacted]`